PROJ OF A GRADED RING IS SEPARATED

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Separatedness criterion. schemes-lemma-characterize-separated (cross-chapter, named). Reduces separatedness to affineness of pairwise intersections and surjectivity of the displayed ring map.

Source reference. Original reference 01KP: focused-seeds/geometry/S9-cons-proj-separated/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a graded ring. The scheme $\mathit{Proj}(S)$ is separated.*

Proof. We have to show that the canonical morphism $\mathrm{Proj}(S) \to \mathrm{Spec}(\mathbf{Z})$ is separated. We will use Schemes, Lemma 01KP. Thus it suffices to show given any pair of standard opens $D_+(f)$ and $D_+(g)$ that $D_+(f) \cap D_+(g) = D_+(fg)$ is affine (clear) and that the ring map

$$S_{(f)} \otimes_{\mathbf{Z}} S_{(g)} \longrightarrow S_{(fg)}$$

is surjective. Any element s in $S_{(fg)}$ is of the form $s = h/(f^n g^m)$ with $h \in S$ homogeneous of degree $n\deg(f) + m\deg(g)$. We may multiply h by a suitable monomial $f^i g^j$ and assume that $n = n'\deg(g)$, and $m = m'\deg(f)$. Then we can rewrite s as $s = h/f^{(n'+m')\deg(g)} \cdot f^{m'\deg(g)}/g^{m'\deg(f)}$. So s is indeed in the image of the displayed arrow. □

Date: stacks-project:constructions.tex:local-snapshot-20261012; GNU FDL 1.2 or later.